2.1 Metric Spaces and Topology: Solutions

Warm-up

1.

(a) Let $\varphi(t) = t/(1+t)$ for $t \geq 0$. Then $\varphi(t) = 0 \iff t = 0$, and φ is increasing since $\varphi(t) = 1 - 1/(1+t)$. It is also subadditive: for $a, b \geq 0$,

$$\varphi(a+b) = \frac{a}{1+a+b} + \frac{b}{1+a+b} \leq \frac{a}{1+a} + \frac{b}{1+b} = \varphi(a) + \varphi(b).$$

Since $d' = \varphi \circ d$, positivity and symmetry are immediate. For the triangle inequality, $d(x,z) \leq d(x,y) + d(y,z)$, and since φ is increasing and subadditive,

$$d'(x,z) \leq \varphi(d(x,y) + d(y,z)) \leq d'(x,y) + d'(y,z).$$

(b) Always $d' \leq d$, so $B_d(x,r) \subseteq B_{d'}(x,r)$. Conversely, if $d'(x,y) = s < 1$ then $d(x,y) = s/(1-s)$. So if $s < 1/2$ then $d(x,y) \leq 2s$, which gives $B_{d'}(x, r/2) \subseteq B_d(x,r)$ for $r \leq 1$. Hence every d-ball around x contains a d'-ball around x and vice versa, so the open sets agree.

(c) $d' < 1$, so $\mathbb{R}$ is bounded under d'. If (x_n) is d'-Cauchy, then for $\varepsilon < 1/2$ there is N with $d'(x_m, x_n) < \varepsilon/2$ for $m, n \geq N$, so $d(x_m, x_n) \leq \varepsilon$. So (x_n) is d-Cauchy, converges in d, and hence in d' (same topology). So $(\mathbb{R}, d')$ is complete.

Remark: so "bounded" is not a topological property, and "closed and bounded ⇒ compact" fails for $(\mathbb{R}, d')$.

2.

For any x, $B(x, 1/2) = \{x\}$, so every singleton is open, so every subset (a union of singletons) is open. Complements of open sets are then all closed, so every subset is closed.

$x_n \to x$ iff for $\varepsilon = 1/2$ there is N with $d(x_n, x) < 1/2$, i.e. $x_n = x$, for $n \geq N$. So the convergent sequences are exactly the eventually constant ones. A Cauchy sequence is, by the same argument with $d(x_m, x_n) < 1/2$, eventually constant, hence convergent: X is complete.

A finite set is always compact. If A is infinite, the cover $\{\{a\} : a \in A\}$ by open sets has no finite subcover. So A is compact iff finite.

3.

(a) The function $y \mapsto d(x,y)$ is continuous (it is 1-Lipschitz by the triangle inequality), so the closed ball $\bar{B} = \{y : d(x,y) \leq r\}$ is closed. It contains $B(x,r)$, so it contains the closure.

(b) Take X with at least two points and the discrete metric, $r = 1$. $B(x,1) = \{x\}$ is closed (Problem 2), so its closure is $\{x\}$, while the closed ball of radius 1 is all of X.

(c) Let $\|y - x\| = r$ (with $r > 0$). For $0 < t < 1$ put $y_t = x + t(y-x)$. Then $\|y_t - x\| = tr < r$, so $y_t \in B(x,r)$, and $\|y_t - y\| = (1-t)r \to 0$ as $t \to 1$. So $y \in \overline{B(x,r)}$. With (a), equality holds.

4.

(⇒) $\overline{f(A)}$ is closed, so $f^{-1}(\overline{f(A)})$ is closed and contains A. Hence it contains $\overline{A}$, i.e. $f(\overline{A}) \subseteq \overline{f(A)}$.

(⇐) Let $F \subseteq Y$ be closed and $A = f^{-1}(F)$. Then $f(\overline{A}) \subseteq \overline{f(A)} \subseteq \overline{F} = F$, so $\overline{A} \subseteq f^{-1}(F) = A$. So A is closed. Preimages of closed sets are closed, so f is continuous.

5.

(a) A Cauchy sequence in A is Cauchy in X, so converges to some $x \in X$. Since A is closed and contains the sequence, $x \in A$.

(b) Let $x \in \overline{A}$ and choose $a_n \in A$ with $a_n \to x$. Then (a_n) is Cauchy, so it converges in A to some $a \in A$. Limits in a metric space are unique, so $x = a \in A$. Hence $\overline{A} = A$.

6.

Finite sets are closed, so points are closed: X is T_1. If U, V are non-empty open sets, $X \setminus U$ and $X \setminus V$ are finite, so $X \setminus (U \cap V)$ is finite; since X is infinite, $U \cap V \neq \emptyset$. So no two points have disjoint neighbourhoods: X is not Hausdorff.

Let $A \subseteq X$ and let $\mathcal{U}$ be an open cover of A. If $A = \emptyset$ there is nothing to do. Otherwise pick $U_0 \in \mathcal{U}$ with $U_0 \cap A \neq \emptyset$; then $U_0 \neq \emptyset$ so $A \setminus U_0 \subseteq X \setminus U_0$ is finite, say $\{a_1, \dots, a_k\}$. Choose $U_i \in \mathcal{U}$ containing a_i. Then $U_0, \dots, U_k$ cover A.

Core

7.

For $n \geq 2$ define f_n to be 0 on $[0, 1/2]$, linear from 0 to 1 on $[1/2, 1/2+1/n]$, and 1 on $[1/2+1/n, 1]$. For $m \geq n$, f_m and f_n agree outside $[1/2, 1/2 + 1/n]$ and $|f_m - f_n| \leq 1$, so $d_1(f_m, f_n) \leq 1/n$. Hence (f_n) is d_1-Cauchy.

Suppose $f_n \to f$ in d_1 with $f \in C[0, 1]$. Then $\int_0^{1/2} |f| \leq d_1(f_n, f) \to 0$, so $\int_0^{1/2} |f| = 0$, and since $|f|$ is continuous and non-negative, $f = 0$ on $[0, 1/2]$. For any $a \in (1/2, 1)$, once $1/n < a - 1/2$ we have $f_n = 1$ on $[a, 1]$, so $\int_a^1 |f - 1| \leq d_1(f_n, f) \to 0$; hence $f = 1$ on $[a, 1]$. As a was arbitrary, $f = 1$ on $(1/2, 1]$. Then f is not continuous at $1/2$, a contradiction. So $(C[0, 1], d_1)$ is not complete.

8.

Work in $X = C[0, 1]$ with d_∞, which is complete. Define

$$(Tf)(x) = e^x + \frac{1}{2} \int_0^1 \cos(xt) f(t)\, dt.$$

Tf is continuous. The function $(x, t) \mapsto \cos(xt)f(t)$ is continuous on the compact square $[0, 1]^2$, hence uniformly continuous. So given $\varepsilon > 0$ there is $\delta > 0$ with $|\cos(xt)f(t) - \cos(x't)f(t)| < \varepsilon$ whenever $|x - x'| < \delta$, for all t. Integrating, the integral term changes by at most $\varepsilon/2$. So $Tf \in C[0, 1]$.

T is a contraction. For $f, g \in X$ and any x,

$$|(Tf)(x) - (Tg)(x)| \leq \frac{1}{2} \int_0^1 |\cos(xt)|\, |f(t) - g(t)|\, dt \leq \frac{1}{2} d_\infty(f, g).$$

So $d_\infty(Tf, Tg) \leq \frac{1}{2} d_\infty(f, g)$. By the contraction mapping theorem T has exactly one fixed point in $C[0, 1]$, which is exactly the statement.

9.

Let X be such a space and suppose X is countable: $X = \{x_1, x_2, \dots\}$ (finite or infinite list). Each $\{x_n\}$ is closed. If $\{x_n\}$ had non-empty interior, it would be open, so x_n would be isolated. So each $\{x_n\}$ is closed and nowhere dense, and X is a countable union of nowhere dense sets. This contradicts the Baire category theorem since X is non-empty and complete.

10.

(a) For $x \in \mathbb{R}$ define the *oscillation*

$$\omega(x) = \inf_{\delta>0} \sup\{|f(y) - f(z)| : y, z \in (x-\delta, x+\delta)\} \in [0, \infty].$$

f is continuous at x iff $\omega(x) = 0$. (If continuous, for $\varepsilon > 0$ choose δ with $|f(y)-f(x)| < \varepsilon/2$ on the interval; then $|f(y) - f(z)| < \varepsilon$. Conversely, $|f(y) - f(x)| \le$ sup over the interval.)

Let $U_n = \{x : \omega(x) < 1/n\}$. U_n is open: if $\omega(x) < 1/n$, choose δ with the sup over $(x-\delta, x+\delta)$ less than $1/n$. For x' in that interval, $(x'-\delta', x'+\delta') \subseteq (x-\delta, x+\delta)$ with $\delta' = \delta - |x - x'|$, so $\omega(x') < 1/n$. Thus the set of continuity points is $\bigcap_n U_n$, a G_δ.

(b) If such f existed, $\mathbb{Q} = \bigcap_n U_n$ would be G_δ. Each $U_n \supseteq \mathbb{Q}$ is open and dense. For each $q \in \mathbb{Q}$, $\mathbb{R} \setminus \{q\}$ is open and dense. These are countably many dense open sets in the complete space $\mathbb{R}$, and their intersection is $\mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \emptyset$, contradicting Baire.

Remark: Thomae's function ($f(p/q) = 1/q$ in lowest terms, $f = 0$ on irrationals) is continuous exactly at the irrationals, so the reverse situation is possible.

11.

Existence. f is continuous (indeed $d(f(x), f(y)) \le d(x, y)$). So $g(x) = d(x, f(x))$ is continuous on the compact space K and attains its minimum at some x_0. If $f(x_0) \neq x_0$ then

$$g(f(x_0)) = d(f(x_0), f(f(x_0))) < d(x_0, f(x_0)) = g(x_0),$$

contradicting minimality. So $f(x_0) = x_0$.

Uniqueness. If $x \neq y$ are both fixed, $d(x, y) = d(f(x), f(y)) < d(x, y)$, which is impossible.

Counterexample. $X = [1, \infty)$ is complete, and $f(x) = x + 1/x$ maps X to X. For $x \neq y$ in X,

$$|f(x) - f(y)| = |x - y| \cdot \left|1 - \frac{1}{xy}\right| < |x - y|,$$

since $xy > 1$ when $x \neq y$ and both are at least 1. But $f(x) = x$ has no solution.

12.

(a) The closed unit ball $\bar{B} = \{x : \|x\| \le 1\}$ is closed (the norm is continuous) and bounded. Let e_n be the sequence with 1 in place n and 0 elsewhere. Then $e_n \in \bar{B}$ and $\|e_m - e_n\| = \sqrt{2}$ for $m \neq n$. No subsequence is Cauchy, so none converges: $\bar{B}$ is not sequentially compact, hence not compact.

(b) Since $\sum 1/n^2 < \infty$, $H \subseteq \ell^2$. Each coordinate map $x \mapsto x_n$ is continuous ($|x_n - y_n| \le \|x - y\|$), so $H = \bigcap_n \{x : |x_n| \le 1/n\}$ is closed in the complete space ℓ^2, hence complete (Problem 5). It remains to show H is totally bounded.

Let $\varepsilon > 0$. Choose N with $\sum_{n>N} 1/n^2 < \varepsilon^2/4$. Let P be the truncation map $P(x) = (x_1, \dots, x_N, 0, 0, \dots)$. $P(H) = \prod_{n \le N}[-1/n, 1/n]$ (padded with zeros) is a closed bounded subset of a copy of $\mathbb{R}^N$, hence compact by Heine–Borel, hence covered by finitely many balls $B(c_j, \varepsilon/2)$. For $x \in H$, pick j with $\|Px - c_j\| < \varepsilon/2$. Then

$$\|x - c_j\| \le \|Px - c_j\| + \|x - Px\| < \frac{\varepsilon}{2} + \Big(\sum_{n>N} \frac{1}{n^2}\Big)^{1/2} < \varepsilon.$$

So H is covered by finitely many ε-balls. Complete and totally bounded, H is compact.

13.

Suppose $h : [0,1] \to S^1$ is a homeomorphism. Then h restricts to a homeomorphism $[0,1] \setminus \{1/2\} \to S^1 \setminus \{h(1/2)\}$. The left side is not connected: it is the disjoint union of the relatively open sets $[0, 1/2)$ and $(1/2, 1]$. The right side is connected: if $h(1/2) = (\cos\theta, \sin\theta)$, then $S^1 \setminus \{h(1/2)\}$ is the image of the connected interval $(\theta, \theta + 2\pi)$ under the continuous map $t \mapsto (\cos t, \sin t)$. Homeomorphisms preserve connectedness, so this is a contradiction.

Similarly, if $h : \mathbb{R} \to \mathbb{R}^2$ were a homeomorphism, $\mathbb{R} \setminus \{0\}$ (disconnected) would be homeomorphic to $\mathbb{R}^2 \setminus \{h(0)\}$, which is path connected (any two points can be joined by at most two line segments avoiding $h(0)$), hence connected. Contradiction.

14.

Connected. S_+ is the image of the connected interval $(0, 1]$ under the continuous map $x \mapsto (x, \sin(1/x))$, so it is connected. Every point $(0, y)$ with $y \in [-1, 1]$ lies in $\overline{S_+}$: choose $\theta \in [-\pi/2, \pi/2]$ with $\sin\theta = y$ and $x_k = 1/(\theta + 2\pi k)$; then $(x_k, \sin(1/x_k)) = (x_k, y) \to (0, y)$. So $S_+ \subseteq S \subseteq \overline{S_+}$. If A is connected and $A \subseteq B \subseteq \overline{A}$, then B is connected. (If $B = U \cup V$ is a separation, A lies in one part, say U; then $V \cap \overline{A} = \emptyset$ because V is relatively open in B and misses A, but $V \subseteq B \subseteq \overline{A}$, so $V = \emptyset$.) So S is connected.

Not path connected. Suppose $\gamma = (x, y) : [0,1] \to S$ is continuous with $\gamma(0) = (0,0)$ and $\gamma(1) = (1, \sin 1)$. The set $Z = \{t : x(t) = 0\}$ is closed and contains 0; let $T = \max Z$. Then $T < 1$ and $x(t) > 0$ for $t > T$. By continuity of y at T, choose $\delta \in (0, 1 - T]$ with $|y(t) - y(T)| < 1/2$ for $t \in [T, T + \delta]$. Let $a = x(T + \delta) > 0$. Choose a positive integer k with $u = 1/(\pi/2 + 2\pi k) < a$ and put $v = 1/(3\pi/2 + 2\pi k) < u$. By the intermediate value theorem applied to x on $[T, T + \delta]$, there are $t_1, t_2 \in (T, T + \delta)$ with $x(t_1) = u$ and $x(t_2) = v$. Since $\gamma(t_i) \in S_+$, $y(t_1) = \sin(1/u) = 1$ and $y(t_2) = \sin(1/v) = -1$. But both must be within $1/2$ of $y(T)$, so $2 = |y(t_1) - y(t_2)| < 1$. Contradiction.

15.

Pick $x_0 \neq x_1$ in X and let $r = d(x_0, x_1) > 0$. The map $g(x) = d(x_0, x)$ is continuous, so $g(X)$ is a connected subset of $\mathbb{R}$, i.e. an interval. It contains 0 and r, so it contains $[0, r]$, which is uncountable. Since g maps X onto a set containing $[0, r]$, X is uncountable.

16.

(a) Each coordinate $\pi_n \circ f$ is the identity $\mathbb{R} \to \mathbb{R}$, which is continuous. By the universal property of the product topology, f is continuous. In the box topology $U = \prod_n (-1/n, 1/n)$ is open, but $f^{-1}(U) = \{t : |t| < 1/n \text{ for all } n\} = \{0\}$, which is not open in $\mathbb{R}$.

(b) A basic open neighbourhood of 0 in the product topology is $W = \prod_k U_k$ with each U_k open, $0 \in U_k$, and $U_k = \mathbb{R}$ for $k > K$. For $n \geq K$, the first n (hence first K) coordinates of $x^{(n)}$ are $0 \in U_k$, and the others lie in $U_k = \mathbb{R}$. So $x^{(n)} \in W$ for $n \geq K$; thus $x^{(n)} \to 0$. In the box topology, $\prod_k (-1/2, 1/2)$ is a neighbourhood of 0 containing no $x^{(n)}$ (each $x^{(n)}$ has some coordinate equal to 1), so $x^{(n)} \not\to 0$.

17.

(a) Let $p : [0,1] \to S^1$, $p(t) = (\cos 2\pi t, \sin 2\pi t)$. It is a continuous surjection, and $p(s) = p(t)$ iff $s = t$ or $\{s,t\} = \{0,1\}$. So p is constant on equivalence classes and, by the universal property of the quotient, induces a continuous bijection $\bar{p} : [0,1]/\sim \to S^1$. The quotient is compact (a continuous image of $[0,1]$ under the quotient map), S^1 is Hausdorff, so $\bar{p}$ is a homeomorphism.

(b) Let $P : [0,1]^2 \to S^1 \times S^1$, $P(s,t) = (p(s), p(t))$. It is continuous and surjective. $P(s,t) = P(s',t')$ iff $p(s) = p(s')$ and $p(t) = p(t')$, i.e. iff ($s = s'$ or $\{s,s'\} = \{0,1\}$) and ($t = t'$ or $\{t,t'\} = \{0,1\}$). This is exactly the equivalence relation generated by the given identifications (the four corners form one class). As in (a), P induces a continuous bijection from the compact quotient to the Hausdorff space $S^1 \times S^1$, hence a homeomorphism.

18.

Let K be compact in the Hausdorff space X and $x \notin K$. For each $k \in K$ choose disjoint open $U_k \ni k$ and $V_k \ni x$. The U_k cover K; choose $k_1, \ldots, k_n$ with $K \subseteq U = \bigcup_i U_{k_i}$. Then $V = \bigcap_i V_{k_i}$ is an open neighbourhood of x disjoint from U, hence from K. So $X \setminus K$ is open.

Example: $\mathbb{R}$ with the cofinite topology. By Problem 6, $[0,1]$ is compact, but it is not closed since the closed sets are $\mathbb{R}$ and finite sets. (Even simpler: any proper non-empty subset of a space with the indiscrete topology.)

Challenge

19.

Fix $\varepsilon > 0$. For $N \geq 1$ let

$$F_N = \{x \in [1,2] : |f(nx)| \leq \varepsilon \text{ for all } n \geq N\} = \bigcap_{n \geq N} \{x \in [1,2] : |f(nx)| \leq \varepsilon\}.$$

Each set in the intersection is closed because $x \mapsto f(nx)$ is continuous, so F_N is closed. By hypothesis, for every x we have $f(nx) \to 0$, so $x \in F_N$ for some N: $[1,2] = \bigcup_N F_N$. Since $[1,2]$ is a complete metric space, Baire gives an N for which F_N has non-empty interior in $[1,2]$, so it contains an open interval (a,b) with $1 \leq a < b \leq 2$.

If $n \geq N$ and $x \in (na, nb)$, then $x/n \in (a,b) \subseteq F_N$, so $|f(x)| = |f(n \cdot x/n)| \leq \varepsilon$. Now choose an integer $M \geq N$ with $M > a/(b-a)$. For $n \geq M$, $n(b-a) > a$, i.e. $(n+1)a < nb$, so consecutive intervals (na, nb) and $((n+1)a, (n+1)b)$ overlap. Since $nb \to \infty$, $\bigcup_{n \geq M} (na, nb) = (Ma, \infty)$. Hence $|f(x)| \leq \varepsilon$ for all $x > Ma$. As ε was arbitrary, $f(x) \to 0$.

20.

f is continuous, so $f(K)$ is compact, hence closed. Suppose $y \notin f(K)$. Then $\varepsilon = \inf\{d(y,z) : z \in f(K)\} > 0$ (a point outside a closed set has positive distance from it). Put $y_n = f^n(y)$ (with

$y_0 = y$). For $n \geq 0$ and $k \geq 1$, since f^n is an isometry,

$$d(y_n, y_{n+k}) = d(y, f^k(y)) \geq \varepsilon,$$

because $f^k(y) \in f(K)$. So any two terms of (y_n) are at distance at least ε, and no subsequence is Cauchy. This contradicts sequential compactness of K. So f is surjective.

21.

Let X be compact Hausdorff. It is T_1 since Hausdorff spaces are.

Step 1 (point vs compact set). By the proof of Problem 18, if K is compact and $x \notin K$, there are disjoint open sets $U \supseteq K$ and $V \ni x$.

Step 2 (closed vs closed). Let A, B be disjoint closed subsets. They are compact (closed subsets of a compact space). For each $a \in A$, Step 1 (with $K = B$, $x = a$) gives disjoint open $U_a \ni a$ and $V_a \supseteq B$. The U_a cover A, so finitely many do: $A \subseteq U = U_{a_1} \cup \cdots \cup U_{a_n}$. Let $V = V_{a_1} \cap \cdots \cap V_{a_n} \supseteq B$, open. If $z \in U \cap V$ then $z \in U_{a_i} \cap V_{a_i}$ for some i, impossible. So U and V separate A and B, and X is normal.

22.

Well defined and surjective: by definition of C, every element is $\varphi(x)$ for some x.

Injective. Let $x \neq y$, and let m be the first index with $x_m \neq y_m$; say $x_m = 1$, $y_m = 0$. Then

$$\varphi(x) - \varphi(y) = 2 \cdot 3^{-m} + \sum_{n>m} 2(x_n - y_n)3^{-n} \geq 2 \cdot 3^{-m} - \sum_{n>m} 2 \cdot 3^{-n} = 2 \cdot 3^{-m} - 3^{-m} > 0.$$

Continuous. Let $\varepsilon > 0$ and choose N with $3^{-N} < \varepsilon$. Given x, the set $W = \{y : y_n = x_n \text{ for } n \leq N\}$ is a basic open set of the product topology containing x. For $y \in W$,

$$|\varphi(x) - \varphi(y)| \leq \sum_{n>N} 2 \cdot 3^{-n} = 3^{-N} < \varepsilon.$$

So φ is continuous at every x (a function between a topological space and a metric space is continuous at x iff each ε-ball around $\varphi(x)$ has a preimage containing a neighbourhood of x).

Homeomorphism. $\{0,1\}^{\mathbb{N}}$ is compact by Tychonoff (each factor is finite, hence compact), and $C \subseteq \mathbb{R}$ is Hausdorff. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.

Remark: this also shows C is compact, uncountable (it is in bijection with $\{0,1\}^{\mathbb{N}}$), and totally disconnected.

23.

(⇒) If X is compact and f continuous, $f(X)$ is a compact subset of $\mathbb{R}$, hence bounded.

(⇐) Suppose X is not compact. Then X is not sequentially compact, so there is a sequence (x_n) with no convergent subsequence. Infinitely many of the x_n are distinct (otherwise some value repeats infinitely often, giving a constant, convergent subsequence), so after passing to a subsequence we may assume the x_n are pairwise distinct. Let $A = \{x_n : n \geq 1\}$.

A has no limit points in X. If p were a limit point of A, every ball $B(p, 1/k)$ would contain infinitely many points of A (in a metric space, a neighbourhood of a limit point meets the set in infinitely many points), so we could choose $n_1 < n_2 < \ldots$ with $x_{n_k} \in B(p, 1/k)$, a subsequence converging to p. Contradiction.

Hence A is closed, and so is every subset of A. So in the subspace topology every subset of A is closed, i.e. A is discrete, and any function on A is continuous. Define $g : A \to \mathbb{R}$ by $g(x_n) = n$. Metric spaces are normal, so by the Tietze extension theorem g extends to a continuous $G : X \to \mathbb{R}$, which is unbounded. This contradicts the hypothesis, so X is compact.